

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)

11
priors compared

32,276
held-out plate appearances

3.05%
held-out HR rate

0.6001
best AUC · KNN comps

served prior

Held-out scores

Fitted on plate appearances before 2026-08-28, scored on the 32,276 that came after (actual HR rate 3.05%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5845	1.53×	0.02950	0.13547	0.00447
KNN comps Prior is the pooled rate of the nearest comparables.	0.6001	1.50×	0.02946	0.13452	0.00426
SVR Support-vector regression on the same feature matrix.	0.5974	1.50×	0.02949	0.13504	0.00614
Random forest Forest regression; a learned similarity metric.	0.5966	1.46×	0.02949	0.13488	0.00607
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6001	1.50×	0.02946	0.13452	0.00426
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6001	1.50×	0.02946	0.13452	0.00426
Linear regression Ordinary least squares on the same features.	0.5963	1.49×	0.02948	0.13482	0.00416
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5973	1.52×	0.02950	0.13495	0.00630
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5964	1.51×	0.02948	0.13482	0.00494
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5966	1.45×	0.02950	0.13494	0.00619

BLEND WEIGHTS (FOREST VARIANT)

KNN 1.00 SVR 0.00 RF 0.00

SHRINKAGE K PER PRIOR

league 248 knn 248 svr 248 rf 248 core 248 ensemble 248
linear 248 logistic 248 nn 248 xgb 248

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,511 PA, actual HR rate 2.36%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5561	1.45×	0.02311	0.11231	0.01006
KNN comps Prior is the pooled rate of the nearest comparables.	0.5834	1.33×	0.02314	0.11213	0.01242
SVR Support-vector regression on the same feature matrix.	0.5842	1.33×	0.02328	0.11503	0.01762
Random forest Forest regression; a learned similarity metric.	0.5926	1.21×	0.02329	0.11334	0.01709
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5834	1.33×	0.02314	0.11213	0.01242
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5834	1.33×	0.02314	0.11213	0.01242
Linear regression Ordinary least squares on the same features.	0.5863	1.08×	0.02322	0.11356	0.01512
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5910	0.96×	0.02323	0.11308	0.01480
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5854	1.33×	0.02318	0.11274	0.01443
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.6006	1.21×	0.02330	0.11334	0.01711

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Ben Rice 21.9%, Munetaka Murakami 19.2%, Luis García Jr. 18.6%, Matt Olson 17.0%, Shohei Ohtani 16.9%, Fernando Tatis Jr. 16.4%
Flat league	5/6	Ben Rice 20.4%, Luis García Jr. 18.0%, Munetaka Murakami 17.6%, Fernando Tatis Jr. 16.3%, Ronald Acuña Jr. 15.6%, Shohei Ohtani 15.3%
KNN comps	6/6	Ben Rice 21.7%, Munetaka Murakami 19.2%, Luis García Jr. 18.9%, Shohei Ohtani 16.9%, Fernando Tatis Jr. 16.7%, Matt Olson 16.5%
SVR	6/6	Ben Rice 22.4%, Munetaka Murakami 19.8%, Luis García Jr. 19.7%, Matt Olson 17.1%, Shohei Ohtani 17.1%, Fernando Tatis Jr. 16.6%
Random forest	5/6	Ben Rice 22.1%, Munetaka Murakami 19.4%, Luis García Jr. 19.1%, Shohei Ohtani 17.2%, Matt Olson 17.0%, Teoscar Hernández 16.3%
Core (KNN+SVR)	6/6	Ben Rice 21.7%, Munetaka Murakami 19.2%, Luis García Jr. 18.9%, Shohei Ohtani 16.9%, Fernando Tatis Jr. 16.7%, Matt Olson 16.5%
Forest blend	6/6	Ben Rice 21.7%, Munetaka Murakami 19.2%, Luis García Jr. 18.9%, Shohei Ohtani 16.9%, Fernando Tatis Jr. 16.7%, Matt Olson 16.5%
Linear regression	6/6	Ben Rice 21.8%, Munetaka Murakami 19.0%, Luis García Jr. 19.0%, Fernando Tatis Jr. 16.9%, Shohei Ohtani 16.8%, Matt Olson 16.7%
Logistic regression	6/6	Ben Rice 22.1%, Luis García Jr. 19.4%, Munetaka Murakami 19.3%, Matt Olson 17.3%, Shohei Ohtani 17.3%, Fernando Tatis Jr. 17.1%
Neural net (torch)	6/6	Ben Rice 22.0%, Luis García Jr. 19.4%, Munetaka Murakami 19.3%, Shohei Ohtani 17.1%, Fernando Tatis Jr. 17.0%, Matt Olson 16.9%
XGBoost	6/6	Ben Rice 22.1%, Munetaka Murakami 19.7%, Luis García Jr. 19.3%, Shohei Ohtani 17.4%, Matt Olson 16.9%, Fernando Tatis Jr. 16.4%

Hitter by hitter

The top 40 bats by the served probability.

MLB Player Performance Analysis - Q3 2023											
HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET	
1Ben RiceNY	slate 21.9%	20.4%	21.7%	22.4%	22.1%	21.7%	21.7%	21.8%	22.1%		
2Munetaka MurakamiMLB	slate 19.2%	17.6%	19.2%	19.8%	19.4%	19.2%	19.2%	19.0%	19.3%		
3Luis García Jr.NYY	18.6%	18.0%	18.9%	19.7%	19.1%	18.9%	18.9%	19.0%	19.4%		
4Matt OlsonML	slate 17.0%	15.3%	16.5%	17.1%	17.0%	16.5%	16.5%	16.7%	17.3%		
5Shohei OhtaniMLB	slate 16.9%	15.3%	16.9%	17.1%	17.2%	16.9%	16.9%	16.8%	17.3%		
6Fernando Tatis Jr.SD	slate 16.4%	16.3%	16.7%	16.6%	16.3%	16.7%	16.7%	16.9%	17.1%		
7Ronald Acuña Jr.ATL	16.3%	15.6%	16.3%	16.3%	16.2%	16.3%	16.3%	16.1%	16.2%		
8Miguel VargasCWS	16.1%	14.6%	16.0%	16.0%	15.8%	16.0%	16.0%	16.0%	16.2%		
9Teoscar HernándezSD	15.7%	15.0%	15.8%	16.0%	16.3%	15.8%	15.8%	16.0%	16.0%		
10Will SmithLAD	15.3%	15.0%	15.2%	15.5%	15.8%	15.2%	15.2%	15.9%	15.7%		
11Jackson ChourioMLB	slate 15.2%	15.2%	15.6%	15.8%	15.9%	15.6%	15.6%	15.7%	15.8%		
12Austin WellsNYY	14.9%	14.7%	15.1%	14.7%	14.7%	15.1%	15.1%	14.9%	14.7%		
13Jackson MerrillSD	14.8%	13.4%	14.7%	14.8%	14.8%	14.7%	14.7%	14.8%	15.2%		
14Spencer JonesNYY	14.6%	13.6%	14.5%	13.4%	13.7%	14.5%	14.5%	13.9%	13.9%		
15Cody BellingerNYY	14.4%	14.6%	14.3%	14.1%	14.5%	14.3%	14.3%	14.4%	14.4%		
16Freddie FreemanMLB	14.4%	13.9%	14.0%	14.0%	13.8%	14.0%	14.0%	14.2%	14.0%		
17Mookie BettsLAD	14.2%	14.2%	14.0%	14.7%	14.1%	14.0%	14.0%	14.4%	14.4%		
18Manny MachadoSD	13.3%	13.0%	13.6%	14.0%	13.7%	13.6%	13.6%	13.8%	13.9%		
19Junior CaminerōTB	13.3%	12.5%	13.1%	14.5%	13.5%	13.1%	13.1%	13.3%	13.7%		
20Kyle TeelCWS	13.2%	13.1%	13.7%	13.0%	13.4%	13.7%	13.7%	13.5%	12.6%		

ML Model Performance Analysis: Seasonal Accuracy (%)											
HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (T)	AVG
21Jonathan Aranda	AB	13.2%	12.4%	13.5%	13.1%	13.2%	13.5%	13.5%	13.2%	13.0%	13.1%
22William Contreras	MIL	13.2%	13.1%	13.3%	13.0%	13.0%	13.3%	13.3%	13.1%	13.1%	13.2%
23Drake Baldwin	ATL	13.0%	12.7%	13.2%	13.5%	13.8%	13.2%	13.2%	13.5%	13.3%	13.2%
24Jo Adell	CLE	12.9%	12.3%	12.7%	13.3%	12.9%	12.7%	12.7%	12.9%	13.2%	12.9%
25George Lombard Jr.	NY	12.9%	13.2%	12.8%	12.5%	13.0%	12.8%	12.8%	13.0%	12.4%	12.8%
26Michael Harris III	ATL	12.7%	12.8%	12.7%	13.2%	12.6%	12.7%	12.7%	12.9%	13.2%	12.8%
27Enrique Hernández	SD	12.6%	12.3%	12.4%	12.1%	12.3%	12.4%	12.4%	12.5%	12.0%	12.4%
28Randal Grichuk	KWS	12.1%	10.9%	12.0%	12.8%	11.9%	12.0%	12.0%	11.9%	12.8%	12.1%
29Austin Riley	ATL	12.0%	11.0%	11.8%	11.7%	11.5%	11.8%	11.8%	11.7%	11.8%	11.8%
30Colson Montgomery	PHI	11.9%	10.8%	12.0%	11.6%	12.0%	12.0%	12.0%	12.3%	12.8%	11.9%
31Jake Bauers	MIL	11.9%	11.0%	12.1%	12.0%	11.9%	12.1%	12.1%	11.9%	11.8%	11.9%
32Ty France	SD	11.8%	11.5%	12.1%	12.0%	12.0%	12.1%	12.1%	12.1%	12.0%	11.9%
33Christian Yelich	MIL	11.6%	11.8%	12.0%	11.3%	11.7%	12.0%	12.0%	12.1%	11.6%	11.8%
34Victor Mesa Jr.	TB	11.5%	11.4%	11.7%	12.0%	11.8%	11.7%	11.7%	11.8%	11.8%	11.7%
35Ozzie Albies	ATL	10.9%	11.3%	10.9%	11.5%	11.5%	10.9%	10.9%	11.2%	11.1%	11.0%
36Andy Pages	LAD	10.6%	10.1%	10.3%	10.0%	10.3%	10.3%	10.3%	10.1%	10.0%	10.2%
37Tristan Peters	CWS	10.5%	11.1%	10.6%	11.0%	11.0%	10.6%	10.6%	10.8%	10.9%	10.7%
38Austin Hays	SD	10.5%	10.1%	9.9%	9.5%	9.5%	9.9%	9.9%	9.9%	9.2%	9.8%
39Mike Yastrzemski	MIL	10.4%	10.4%	10.4%	10.3%	10.3%	10.4%	10.4%	10.1%	10.1%	10.3%
40Kyle Tucker	LAD	10.4%	10.5%	10.4%	10.3%	10.4%	10.4%	10.4%	10.3%	10.1%	10.3%

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

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